# Khalil Krifi

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## EXPERIENCE

**Backend Engineering Intern: Natural Language Query Interface for Logs & Metrics** Feb 2026 – Jun 2026
*Cognira*, Tunis, Tunisia (On-site) (Repository)

- Designed an event-driven backend with Scala, Python, and Kafka to translate natural-language requests into OpenSearch DSL and PromQL queries.
- Built a schema-aware LangGraph pipeline to validate and self-correct generated queries before execution, cutting invalid query executions from 20% to 7% on a 100-query evaluation set.
- Implemented Redis-backed semantic caching with vector embeddings, reaching an 80% cache hit rate and reducing repeat-query latency from 3.4 s to 200 ms; added multi-tenant RBAC, TOTP MFA, and encrypted API-key storage.

**Software Engineer Intern: Fleet Management System** Jul 2025 – Sep 2025
*Omnilink*, Tunis, Tunisia (Remote) (Repository)

- Designed a Spring Boot microservices backend split into independently deployable services, separating business domains to support independent deployment and scaling.
- Developed 40+ GraphQL APIs with Apollo and Netflix DGS for vehicle telemetry, driver allocation, and trip management.

**Backend Engineering Intern: Falsified Signature Detection** Feb 2025 – May 2025
*Laval University*, Quebec City, Canada (Remote) (Repository)

- Built a Spring Boot microservices backend for user management, image analysis, and reporting, using Spring Cloud Gateway, Eureka, and centralized configuration.
- Developed an asynchronous analysis workflow integrating PyTorch inference with Kafka-based report generation and email notifications, keeping API response time under 150 ms.
- Implemented secure evidence storage with MongoDB GridFS, EXIF extraction, and chain-of-custody auditing.

**Software Engineer Intern: Enterprise Content Management** Jun 2024 – Aug 2024
*BFI Group*, Tunis, Tunisia (On-site) (Repository)

- Developed backend services for enterprise document management, supporting hierarchical storage, metadata tracking, versioning, and files up to 500 MB.
- Indexed 2,000+ documents with Elasticsearch, achieving average retrieval latency below 300 ms; implemented JWT authentication and granular role-based access control.

## PROJECTS

**DevForge: Internal Developer Platform** Spring Boot, Go, Kafka, Keycloak, PostgreSQL (Repository)

- Built a Spring Boot provisioning API with 13 versioned templates, dynamic placeholder resolution, and automated GitHub repository creation with preconfigured CI files, reducing project setup from roughly 15 min to under 20 s.
- Implemented asynchronous service provisioning through Kafka and a dedicated Go worker, decoupling long-running generation tasks from API requests.

**Etude.AI: Microservice Study Platform** *(Team Project)* Java, Spring Boot, PostgreSQL, FastAPI (Repository)

- Developed Java 21/Spring Boot services for learning sessions and gamification, using PostgreSQL, Flyway migrations, Redis caching, and Keycloak JWT authentication.
- Integrated a FastAPI LLM service through reactive WebClient, with circuit breakers, exponential-backoff retries, and gateway timeouts.

## EDUCATION

**National Engineering Degree in Computer Science** Sep 2023 – Jun 2026
École Nationale des Sciences de l'Informatique (ENSI), University of Manouba, Tunis, Tunisia

**Engineering Preparatory Studies (Mathematics & Physics)** Sep 2021 – Jun 2023
Institut Préparatoire aux Études d'Ingénieurs de Nabeul (IPEIN), Nabeul, Tunisia

## SKILLS

**Languages:** Java, SQL, Scala, Go, Python, TypeScript, C/C++
**Backend & Testing :** Spring Boot, Spring Cloud, Microservices, REST APIs, GraphQL, Locust, K6
**Databases & Messaging:** PostgreSQL, Redis, MongoDB, Elasticsearch, Apache Kafka
**Infrastructure & Observability:** Git, Docker, Kubernetes, GitHub Actions, ArgoCD, Prometheus, Grafana
**Languages Spoken:** English (C1), French (B2), German (A2)